

The type of values are fixed or variable.

· **NUMERIC:**

These datatypes can include exact numeric datatypes

For ex: integer, float, bits

DATATYPE NAMES	SIGNED	UNSIGNED	STORAGE
TINYINT (very small integer)	-128 to 127	0 to 255	1 Byte
SMALLINT	-32768 to 32767	0 to 65535	2 Bytes
MEDIUMINT	-8388608 to 8388607	0 to 16777215	3 Bytes
INT	-2147483648 to 2147483647	0 to 4294967295	4 Bytes
BIGINT	- 92233720368547758 08 to 9223372036854775807	0 to 1844674407370955161 5	8 Bytes

· **FLOAT(m,d):**

It is a floating-point number that cannot be unsigned. It defines the display length(m) and the number of decimals(d).

Example:

FLOAT(5,2)

You can store numbers like:

123.45 → total 5 digits, 2 after decimal

· **DOUBLE(m,d):**

In this datatype decimal decision can go to 53 places and it requires 8 Bytes for storage. It cannot be unsigned.

DOUBLE(8,3)

Can store values like:

12345.678 → total 8 digits, 3 after decimal

· **DECIMAL(m,d):**

In this datatype each decimal corresponds to byte.

Example:

DECIMAL(7,2)

Can store values like:

12345.67

· **BIT(m):**

It is used to store the bit values into the table column.

Here m defines number of bit per value that has a range of 1 to 64.

· **BOOL:**

It is used only for the true and false condition. It is considered numeric value 1 as True and 0 as False.

· **BOOLEAN:**

Similar to BOOL.

FLOAT	DOUBLE	DECIMAL
Approximate value	More precise approximate value	Exact value
Uses 4 bytes	Uses 8 bytes	Size depends on m,d
Good for scientific data	Good for high-precision calculations	Best for money & financial data
Not fully accurate	More accurate than float	100% accurate

Summary Table

Type	Default m,d
FLOAT	No default
DOUBLE	No default
DECIMAL	(10,0)

· **STRING DATATYPES**

The string datatypes is used to hold plain text and binary data.

NAME	MAX SIZE	FIXED/VARIABLE
CHAR(size)	255 characters	Fixed length string
VARCHAR(size)	255 characters	Variable length string
TINYTEXT(size)	255 characters	Number of characters to store
TEXT(size)	65535	Number of characters to store
MEDIUM(size)	16777215	
LONGTEXT(size)	4GB or 4294967295	
BINARY(size)	255 characters	Number of binary characters stored or fixed length string
VARBINARY(size)	255 characters	Variable length string

- **BINARY LARGE OBJECT DATATYPE (BLOB):** These are **BLOB (Binary Large Object) data types** used in databases like MySQL to store large binary data (e.g., images, videos, audio files, or documents).

TINYBLOB	It can hold a max size of 255 bytes
BLOB	It can hold a max size of 65535 bytes
MEDIUMBLOB	It can hold a max size of 16777215 bytes
LONG	It can hold a max size of 4GB/ 4294967295 bytes

- **DATE AND TIME DATATYPES**

This datatype is used for temporal values such as date, time, datetime, timestamp and year

DATATYPE	RANGE/SIZE	STORAGE	FORMAT
Year[(2/4)]	2 digits or 4 digits	1 Byte	YYYY
DATE	1000-01-01 to 9999-12-31	3 Bytes	YYYY-MM-DD
TIME	00:00:00 to 838:59:59	3 Bytes	HH:MM:SS
DATETIME	1000-01-01 00:00:00 to 9999-12-31 23:59:59	5 Bytes	YYYY-MM-DD HH:MM:SS
TIMESTAMP(m)	1970-01-01 00:00:01 to 2038-01-19 03:14:07	4 Bytes	YYYY-MM-DD HH:MM:SS

In TIMESTAMP(m), the m specifies how many digits to store after the seconds (fractional seconds).

Example:

TIMESTAMP(3)

This will store time like:

2025-01-01 10:20:30.123 → (3 digits = milliseconds)

TIMESTAMP(6)

This will store:

2025-01-01 10:20:30.123456 → (6 digits = microseconds)